

**Note for (b):** If the domain consists of exactly one element, it's easy to see from the definition of universal and uniqueness quantifier why  $\exists! x P(x)$  is True if  $\forall x P(x)$  is True. By defn. of conditional,  $\boxed{\forall x P(x) \rightarrow \exists! x P(x) \text{ is True.}}$

56) Write out  $\exists! x P(x)$ , where the domain consists of the integers 1, 2 and 3 in terms of negations, conjunctions and disjunctions.

**Soln:** **Claim:**  $\exists! x P(x)$  and  $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$  are logically equivalent.

**Proof:** Let,  $\exists! x P(x)$  be True. Given, Domain =  $\{1, 2, 3\}$ . By definition of uniqueness quantifier, exactly one of  $P(1)$ ,  $P(2)$  or  $P(3)$  is True. Wlog, let  $P(1)$  be True.  $\therefore P(2)$  and  $P(3)$  are both False. By defn. of negation,  $\neg P(2)$  and  $\neg P(3)$  are both True. By defn. of AND,  $\boxed{P(1) \wedge \neg P(2) \wedge \neg P(3)}$  is True. By defn. of OR,  $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$  is True.

Let  $P(2)$  be True.  $\therefore P(1)$  and  $P(3)$  are both False. By defn. of negation,  $\neg P(1)$  and  $\neg P(3)$  are both True. By defn. of AND,  $P(2) \wedge \neg P(1) \wedge \neg P(3)$  is True. By defn. of OR,  $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$  is True.

Let  $P(3)$  be True.  $\therefore P(1)$  and  $P(2)$  are both False. By defn. of negation,  $\neg P(1)$  and  $\neg P(2)$  are both True. By defn. of AND,  $P(3) \wedge \neg P(1) \wedge \neg P(2)$  is True. By defn. of OR,  $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$  is True.

$\boxed{\exists! x P(x) \text{ is True} \Rightarrow (P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2)) \text{ is True}}$